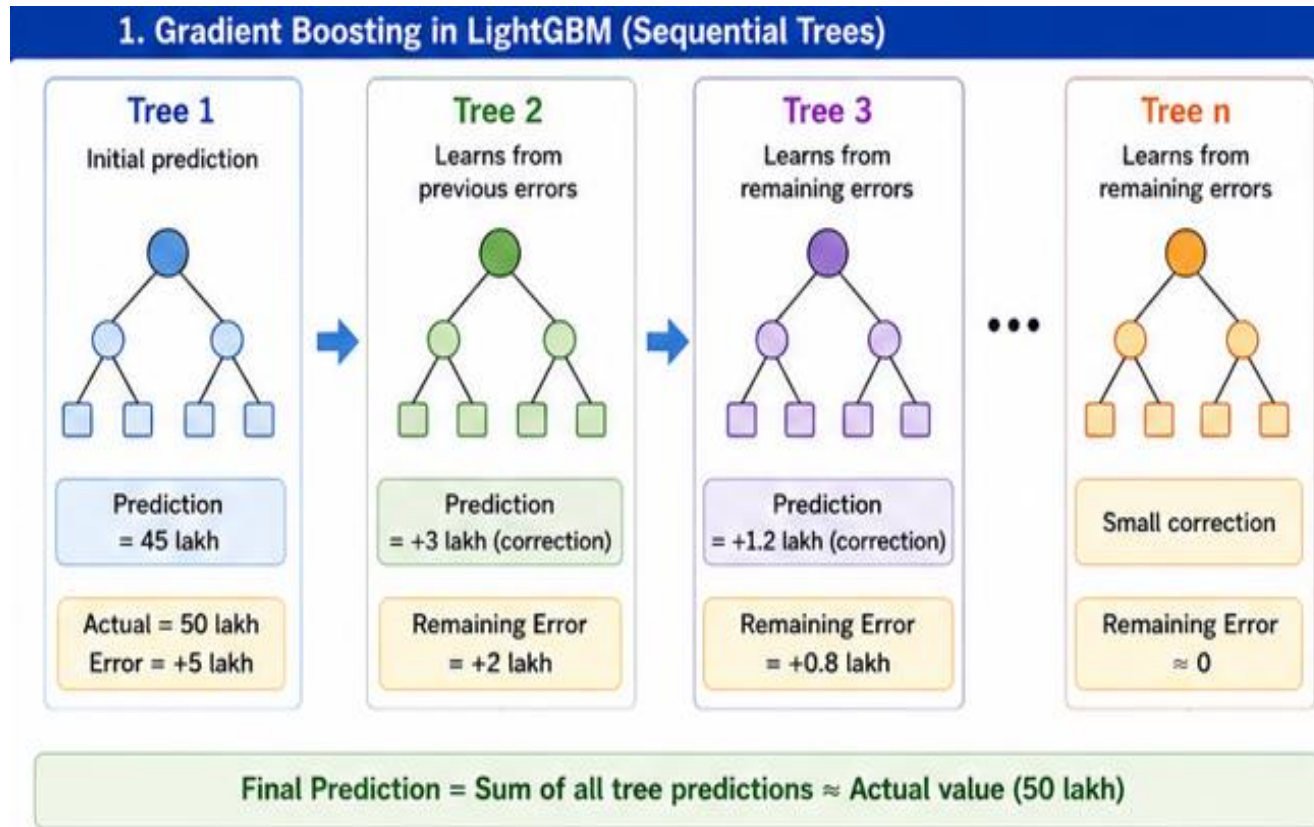


LightGBM

- **LightGBM (Light Gradient Boosting Machine)** is a machine learning algorithm used for **regression** (predicting numbers) and **classification** (predicting categories).
- It builds **many small decision trees**, where **each new tree learns from the mistakes (errors) of the previous trees**.

SEQUENTIAL TREE



LOSS FUNCTION

LightGBM uses loss functions to minimize the difference between actual (y) and predicted ($\hat{y}$)

a) Mean Squared Error (MSE)

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Actual (y)	Predicted ($\hat{y}$)	Error ($y - \hat{y}$)	Error ²
50	47	3	9
60	58	2	4
70	65	5	25

$$\text{MSE} = (9 + 4 + 25) / 3 = 12.67$$

- Squares the errors
- Gives more weight to larger errors
- Sensitive to outliers

b) Mean Absolute Error (MAE)

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

Actual (y)	Predicted ($\hat{y}$)	Absolute Error $ y - \hat{y} $
50	47	3
60	58	2
70	65	5

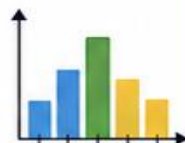
$$\text{MAE} = (9 + 2 + 5) / 3 = 3.33$$

- Uses absolute errors
- Treats all errors equally
- Less affected by outliers than MSE

FEATURES

1 Histogram-Based Learning

Continuous numerical values are converted into histogram bins (intervals). Instead of checking every unique value, it works with bins to find the best split.



Example:

Original values (ages)

21	22	23	24	25	26	27	28	29
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Histogram bins

20 – 24 → Bin 1	25 – 29 → Bin 2
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Benefits

- Fewer values to evaluate
- Faster split finding
- Lower memory usage

3 Exclusive Feature Bundling (EFB)

If two or more features are rarely non-zero at the same time, they are bundled into a single feature.

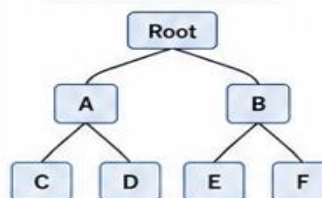
Example:

	Customer	Student	Retired		Bundled Feature
ID 1	1	1	0	Bundled	1
ID 2	1	0	1		2
ID 3	1	1	0		1
ID 4	1	0	1		2

2 Leaf-Wise Tree Growth

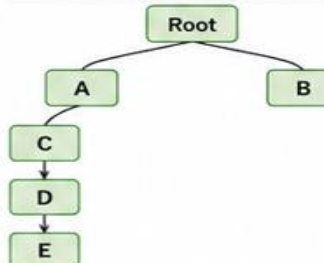
LightGBM grows the leaf with the highest gain (maximum reduction in loss). It does not grow level by level.

Level-wise growth



- Grows level by level
- May need more trees

Leaf-wise growth (LightGBM)



- Splits the leaf with highest gain
- Deeper trees on promising paths
- Better accuracy with fewer trees
- Can overfit on small datasets

4 Gradient-based One-Side Sampling (GOSS)

After each tree, some samples have large gradients (errors), while others have small gradients.

- Keeps all samples with large gradients
- Randomly samples only some of the small-gradient samples

Example:

Sample	Gradient (Error)		Used by LightGBM
A	10.0	→	Keep (Large gradient)
B	8.0	→	Keep (Large gradient)
C	0.3	→	Sample (Small gradient)
D	0.1	→	Sample (Small gradient)
...	...	→	...

SUMMARY

Summary of Key Techniques			
Technique	What it does	How it works	Benefit
 Histogram-Based Learning	Convert continuous values into bins (histograms)	Groups values into fixed bins and uses them to find best splits	Faster training, less memory
 Leaf-Wise Tree Growth	Splits the leaf with the highest gain	Always chooses the leaf that gives maximum loss reduction	Higher accuracy, faster convergence
 Exclusive Feature Bundling (EFB)	Bundles mutually exclusive sparse features	Combines features that are rarely non-zero together	Reduces feature dimension, lower memory usage
 Gradient-based One-Side Sampling (GOSS)	Focuses on samples with large gradients (errors)	Keeps all large-gradient samples and randomly samples small-gradient ones	Faster training while maintaining accuracy

WORKFLOW

